In the figure 13.1 huge and huge cables are used during network cabling. This Causes lot of confusion which cable to be connected and therefore alternatives to replace this are in process.

13.2 Types of Network Cables

There are many types of cables available in the networking. Here we are going to discuss about six different types of cables.

1. **Coaxial Cables:** This cable was invented at late 1880's, which is used to connect the television sets to home antennas. This cable is used to transfer the information in 10 mbps. The cable is divided into thinnet and thicknet cables. These cables have a copper wire inside and insulation is covered on the top of the copper wire to provide protection to the cable. These cables are very difficult to install and maintain, because they are too big to carry and replace. The coaxial cable got its name by the word "coax". Nowadays coaxial cables are also used for dish TV where the setup box and the television is connected using the coaxial cable only. Some of the cable names are Media Bridge 50-feet Coaxial cable, Amazon basics CL2-Rated Coaxial cables, etc. See Figure 13.2



Figure13.2 Coaxial cables for connecting television sets

2. **Twisted Pair Cables:** It is type of cable with two or more insulated wires twisted together. It started with a speed of 10 mbps (10BASE-T cable is used). Then the cable was improved and the speed was higher and went to 100 mbps and the cable was renamed as 100BASE-TX. Then finally the cable was improved to 10 gbps and named as 10GBASE-T. This twisted cable has 8 wires which are twisted to ignore electromagnetic interference. The eight wires cannot be placed in a single unit as there can be difficulty in space, so it is twisted to make it one wire. There are two types of twisted pair cables, Unshielded Twisted Pair (UTP) and Shielded Twisted pair (STP). The UTP is used nowadays as modern cables for Internet and they are lower in cost and installation and maintenance is easy compared to the coaxial cables. STP is similar to UTP, but it is covered by an additional jacket to protect the wires from External interference. See Figure 13.3

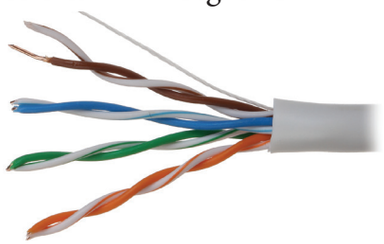


Figure13.3 Twisted pair cables

3. **Fiber Optics:** This cable is different from the other two cables. The other two cables had an insulating material outside and the conducting material like copper inside. But in this cable it is strands of glass and pulse of light is used to send the information. They are mainly used in Wide Area Network (WAN). The WAN is a

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